21-242 Recitation 12

Fundamental theorems — Q1. State the fundamental theorems of linear algebra

Course-level solution, with a later-course alternative when useful

October 1, 2026

Prompt

State the fundamental theorems of linear algebra.

Course-level solution

At the level of this midterm (finite-dimensional spaces, no inner products required), the core theorems are the following.

Theorem (Rank–nullity)**.** If $T : V \to W$ is linear and $\dim V < \infty$, then $\dim \operatorname{im}(T) + \dim \ker(T) = \dim V$.

Theorem (Rank)**.** For $A \in \mathbb{F}^{m \times n}$,

$$\operatorname{rank}(A) := \dim \operatorname{im}(T_A) = (\text{number of pivots in } \operatorname{RREF}(A))$$

equals the dimension of the column space and of the row space of A.

Theorem (Characterizations of injectivity / surjectivity)**.** Let $T : V \to W$ with $\dim V = n < \infty$, $\dim W = m < \infty$, and let $\mathcal{B}$ be a basis of V.

- T injective $\iff \ker T = \{\vec{0}\} \iff T(\mathcal{B})$ independent $\iff \dim \operatorname{im}(T) = n \iff n \leq m$ and rank $= n$.
- T surjective $\iff \operatorname{im}(T) = W \iff T(\mathcal{B})$ spans $W \iff \dim \operatorname{im}(T) = m \iff m \leq n$ and rank $= m$.
- If $n = m$, these are all equivalent to T being invertible.

For T_A, injectivity means every column is a pivot; surjectivity means every row has a pivot.

Theorem (Left and right inverses)**.** T_A is injective iff A has a left inverse ($BA = I_n$ for some B). T_A is surjective iff A has a right inverse ($AC = I_m$ for some C).

(If the course has already named the "four subspaces" theorem over $\mathbb{R}$ with orthogonality, that is a later refinement using a transpose/inner product and is not needed for the proofs below.)